

LEVEL 23 — NumPy Date and Time

Easy, line-by-line examples with output

1. np.datetime64 — Create a Date

```
import numpy as np

date = np.datetime64("2026-08-26")
print(date)
```

Output:
2026-08-26

2. np.datetime64 — Create Date and Time

```
import numpy as np

dt = np.datetime64("2026-08-26T18:30")
print(dt)
```

Output:
2026-08-26T18:30

3. Date Creation

```
import numpy as np

start = np.datetime64("2026-01-01")
print(start)
```

Output:
2026-01-01

4. Date Arithmetic — Add Days

```
import numpy as np

date = np.datetime64("2026-08-26")
new_date = date + np.timedelta64(5, "D")

print(new_date)
```

Output:
2026-08-31

5. Date Arithmetic — Subtract Days

```
import numpy as np

date = np.datetime64("2026-08-26")
new_date = date - np.timedelta64(10, "D")

print(new_date)
```

Output:
2026-08-16

6. np.timedelta64 — Store a Time Difference

```
import numpy as np

days = np.timedelta64(7, "D")
print(days)
```

Output:
7 days

7. Date Differences

```
import numpy as np

date1 = np.datetime64("2026-08-01")
date2 = np.datetime64("2026-08-10")

difference = date2 - date1
print(difference)
```

Output:
9 days

8. Date Ranges — np.arange()

```
import numpy as np

dates = np.arange(
    np.datetime64("2026-08-01"),
    np.datetime64("2026-08-06")
)

print(dates)

Output:
['2026-08-01' '2026-08-02' '2026-08-03'
 '2026-08-04' '2026-08-05']
```

9. Date Ranges — Every 2 Days

```
import numpy as np

dates = np.arange(
    np.datetime64("2026-08-01"),
    np.datetime64("2026-08-11"),
    np.timedelta64(2, "D")
)

print(dates)

Output:
['2026-08-01' '2026-08-03' '2026-08-05'
 '2026-08-07' '2026-08-09']
```

10. Business Days

```
import numpy as np

dates = np.arange(
    np.datetime64("2026-08-24"),
    np.datetime64("2026-08-29"),
    dtype="datetime64[D]"
)

print(dates)
print(np.is_busday(dates))

Output:
['2026-08-24' '2026-08-25' '2026-08-26'
 '2026-08-27' '2026-08-28']
[ True  True  True  True  True ]
```

Note:

These dates are Monday to Friday, so all are business days.

11. np.is_busday() — Check a Date

```
import numpy as np

date = np.datetime64("2026-08-29")

print(np.is_busday(date))
```

Output:
False

Reason:
2026-08-29 is Saturday.

12. np.busday_count() — Count Business Days

```
import numpy as np

start = np.datetime64("2026-08-24")
end = np.datetime64("2026-08-29")

count = np.busday_count(start, end)

print(count)
```

Output:
5

Note:
The end date is not included.
Monday to Friday = 5 business days.

13. np.busday_count() — Simple Example

```
import numpy as np

start = np.datetime64("2026-08-01")
end = np.datetime64("2026-08-10")

print(np.busday_count(start, end))
```

Output:
5

Note:
Only Monday-Friday are counted.

14. Complete Example — Date Difference + Business Days

```
import numpy as np

start = np.datetime64("2026-08-24")
end = np.datetime64("2026-09-01")

calendar_days = end - start
business_days = np.busday_count(start, end)

print("Calendar days:", calendar_days)
print("Business days:", business_days)
```

Output:
Calendar days: 8 days
Business days: 6

15. Quick Revision

<code>np.datetime64()</code>	-> Creates dates and times
<code>np.timedelta64()</code>	-> Represents a time difference
<code>date + timedelta</code>	-> Adds time
<code>date - timedelta</code>	-> Subtracts time
<code>date2 - date1</code>	-> Finds date difference
<code>np.arange()</code>	-> Creates date ranges
<code>np.is_busday()</code>	-> Checks business day
<code>np.busday_count()</code>	-> Counts business days